

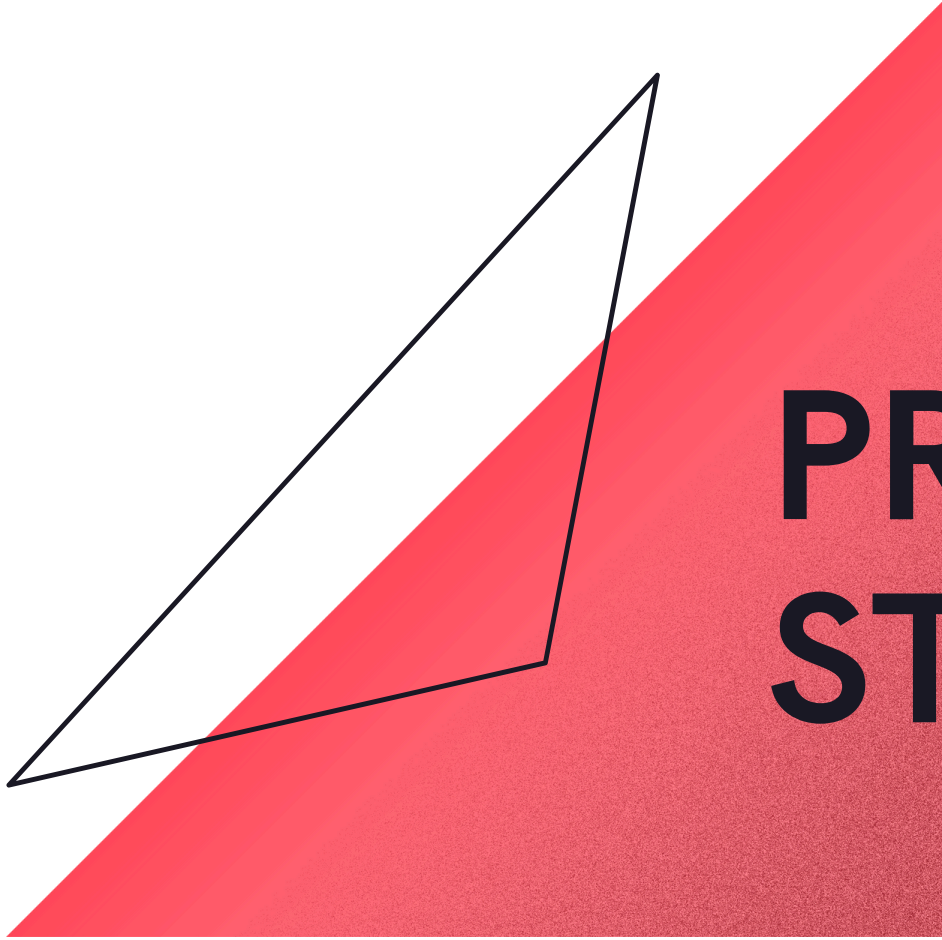


Latent Diffusion



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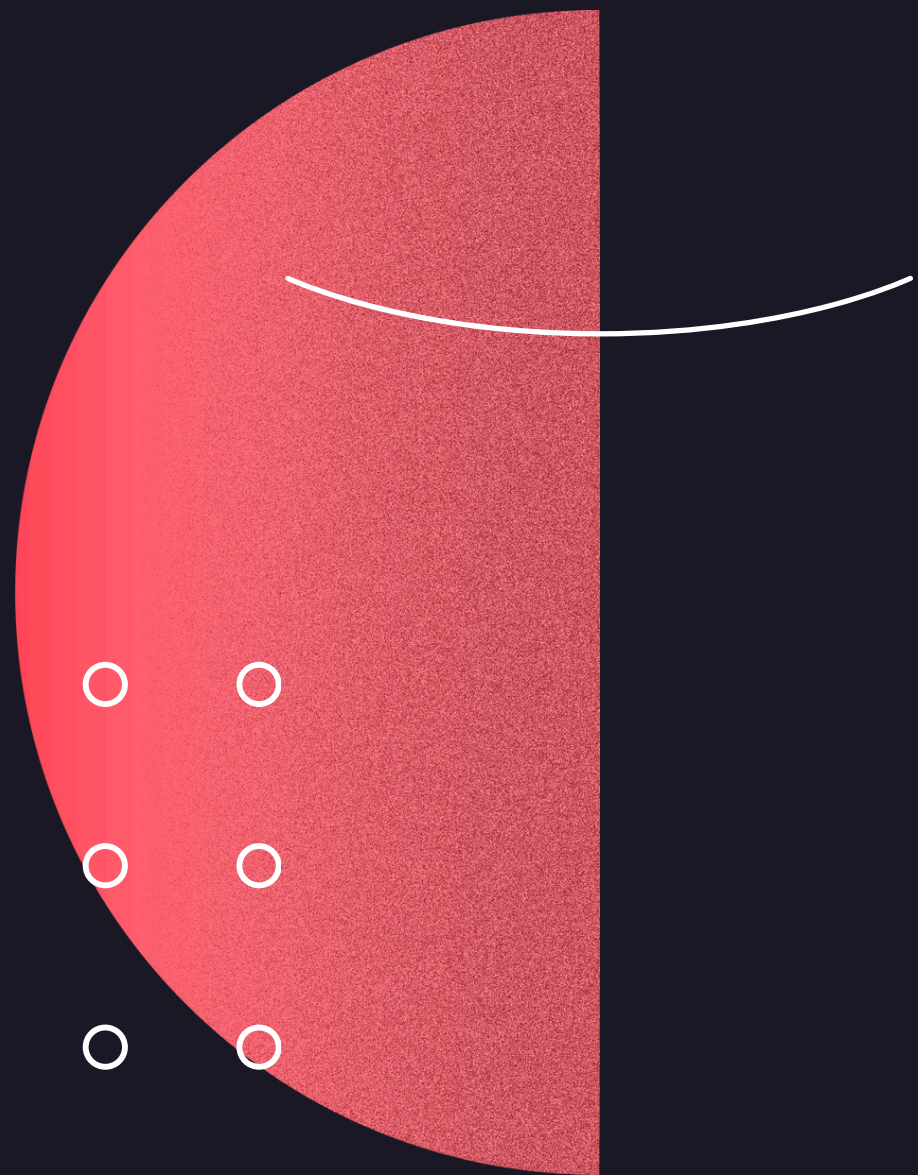
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PROBLEM STATEMENT

Build a full-stack text-to-image Generative AI web application using the Stable Diffusion architecture, where only the U-Net diffusion model is fine-tuned on a custom image-caption dataset while keeping the pre-trained CLIP tokenizer and VAE encoder-decoder components frozen

DATASET



1. Contains 1,200+ Naruto-style images sourced from narutopedia.com (700 MB)
2. Each image is paired with a BLIP-generated caption (text + PIL JPEG)
3. Provided as a train-only split, ideal for fine-tuning diffusion models
4. Captions include stylized prompts like "Banksy street art of ninja"
5. <https://huggingface.co/datasets/lambdalabs/naruto-blip-captions>



Architecture

CLIP

CLIP tokenizer converts text into embeddings that capture semantic relationships between text and images, enabling effective text-to-image generation.

ENCODER

The VAE encoder compresses images into a latent space, facilitating efficient image manipulation and generation.

DECODER

The VAE decoder reconstructs images from latent representations, enabling new image generation based on compressed data.

UNET

U-Net, through iterative denoising, is used in diffusion models to generate high-quality images from random noise.

FINE-TUNING

Fine-tuning the diffusion model with an image-caption dataset improves the alignment of generated images with textual prompts while keeping encoder weights frozen.

WHAT IS LATENT DIFFUSION ?



Latent Diffusion is a technique that generates images by operating in a compressed latent space instead of pixel space, making it more efficient and scalable.

Put Simply : It's like sketching a rough blueprint (latent space) before painting the full artwork — faster and easier than working on the full canvas from the start.

OUR PRE-TRAINED MODELS

Stable diffusion v1.5 :

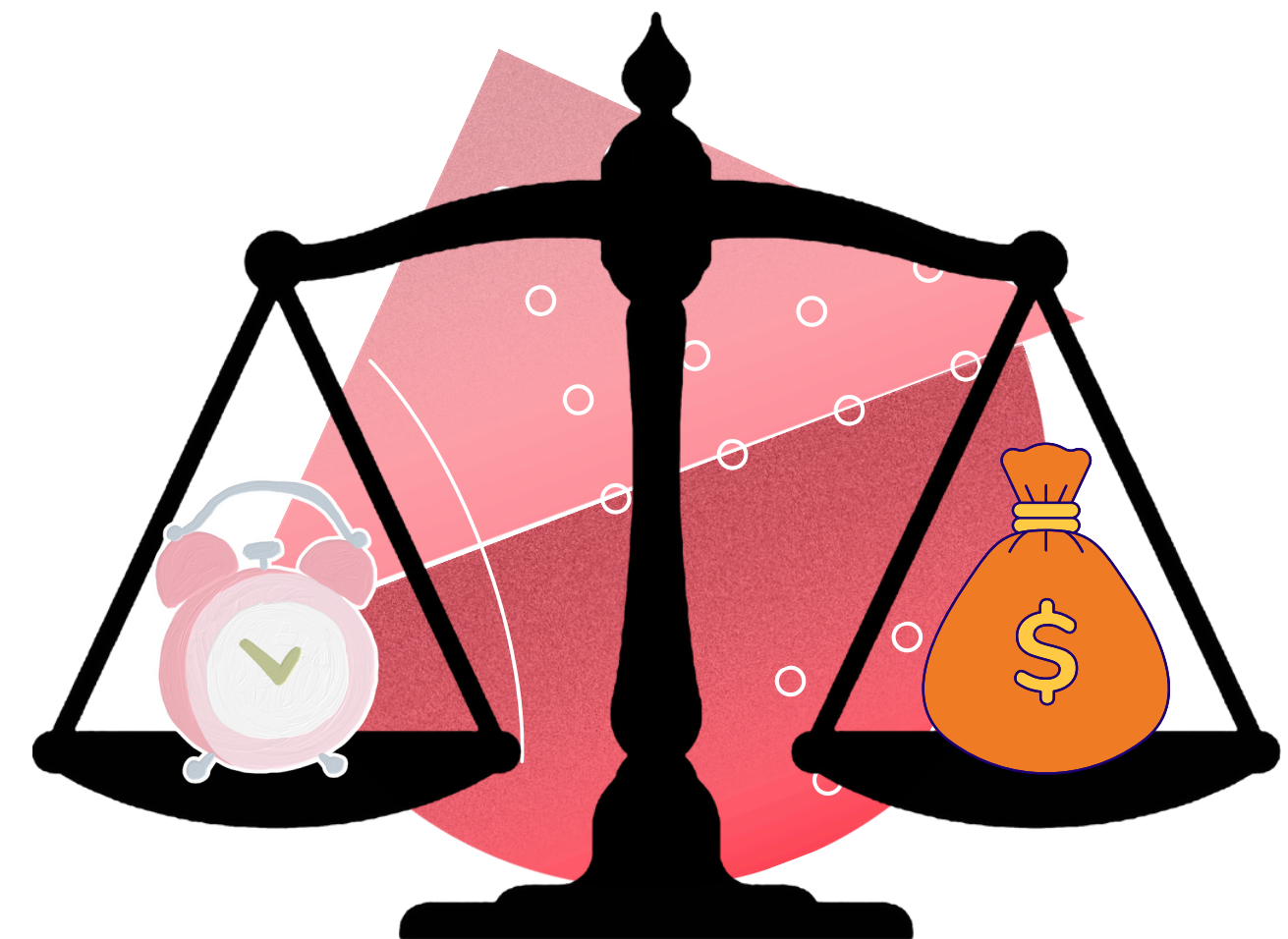
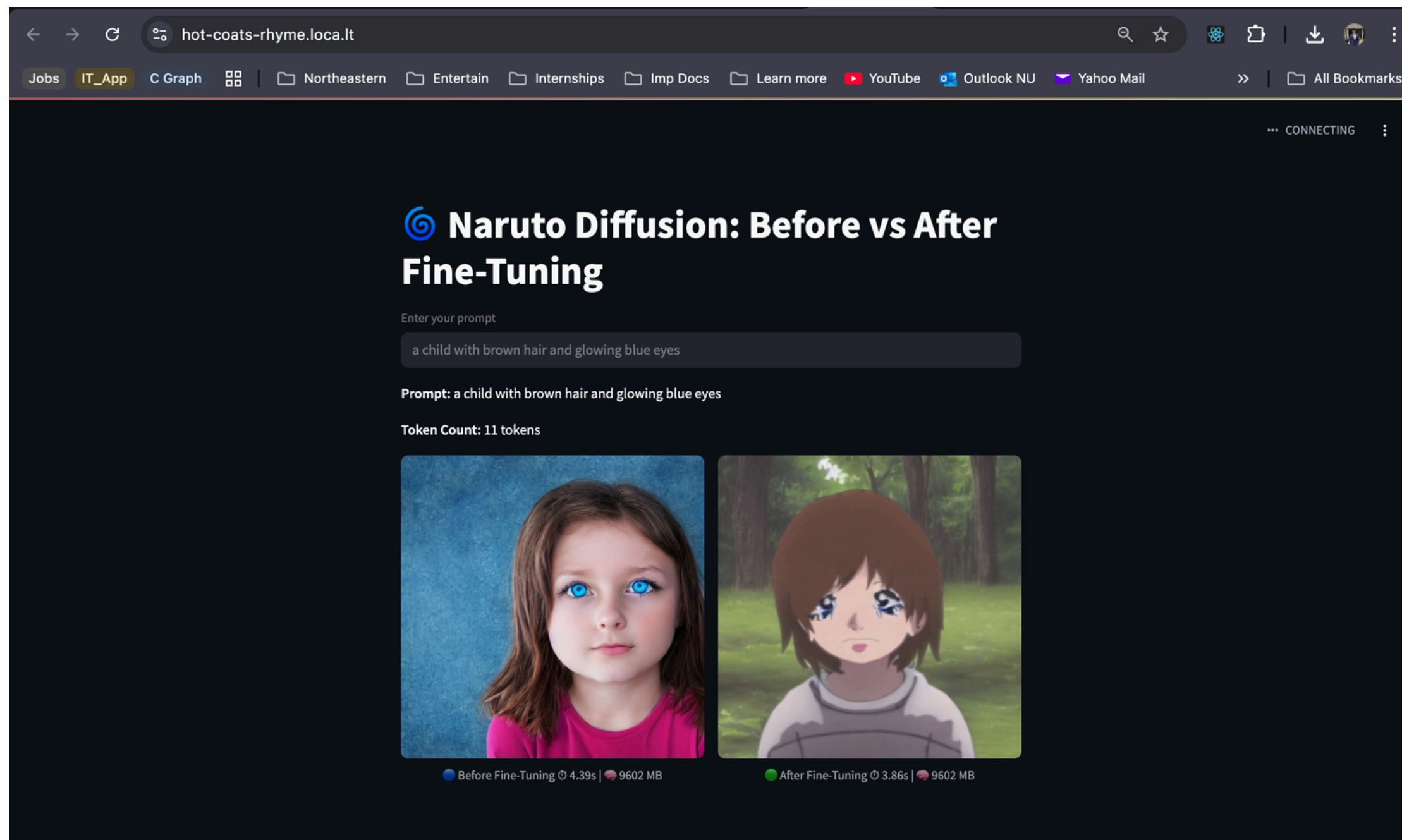
1. A powerful text-to-image LDM by Stability AI.
2. Trained on millions of image-text pairs to create realistic or artistic images from any prompt.
3. Uses a smart text reader (CLIP) and a noise-removing model (U-Net).
4. Has around 1.25 billion parameters.
5. Best use cases include Art, design, fantasy scenes, concepts

Mo-Di-Diffusion:

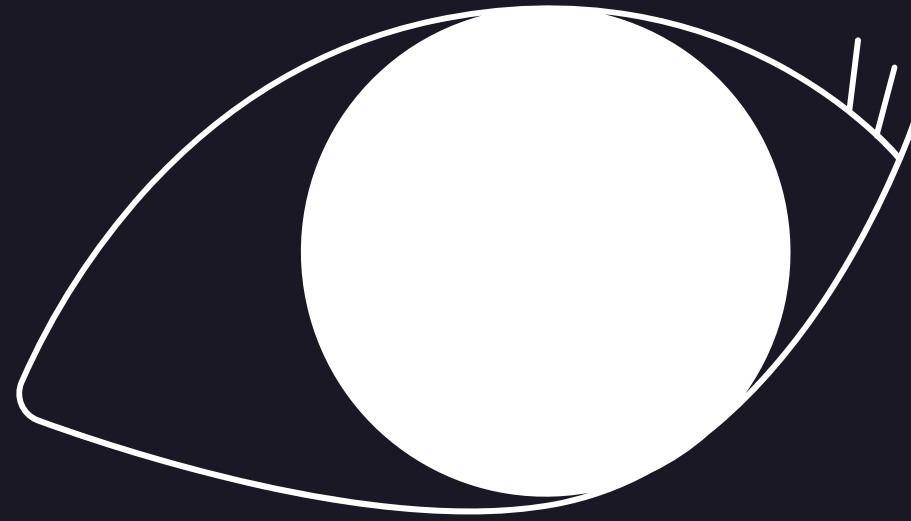
1. A fine-tuned version of Stable Diffusion made by Nitrosocket.
2. Specially trained to create images in a “modern Disney” style.
3. Has 1.25B parameters
4. Trained using DreamBooth with 9,000 steps.
5. Works well for cartoon-like characters, animals and scenes.



STREAMLIT UI



Combination of Backend + Frontend + Deployment



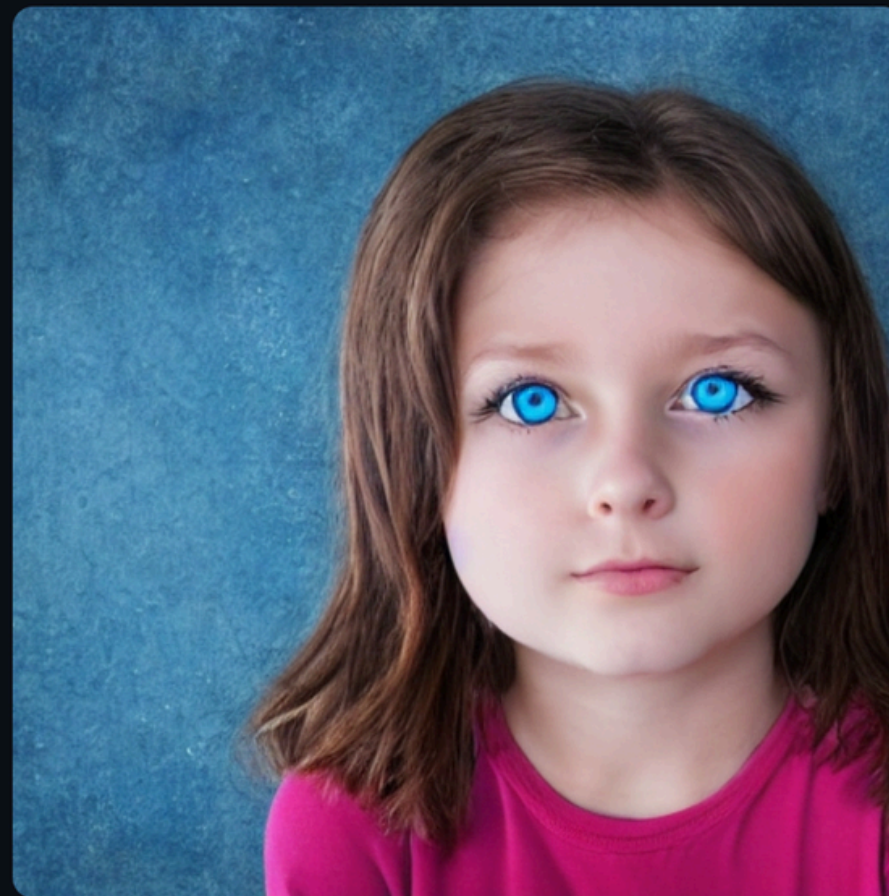
HERE ARE SOME PREDICTIONS

STABLE DIFFUSION V1-5

10 Epochs

MO DI DIFFUSION

5 Epochs



Before Fine-Tuning ⌚ 4.39s | 🧠 9602 MB



After Fine-Tuning ⌚ 3.86s | 🧠 9602 MB



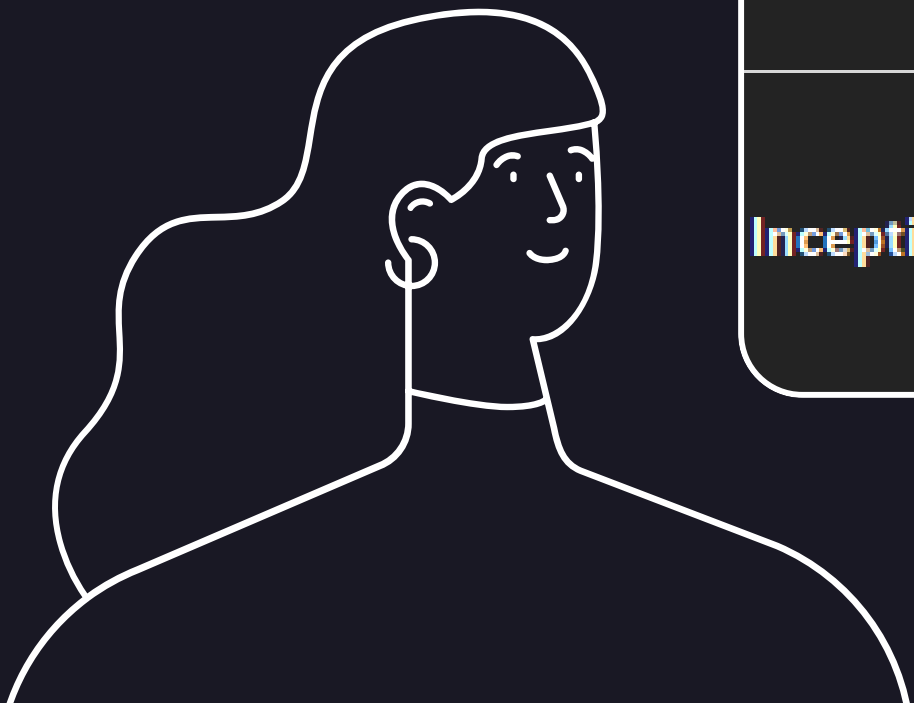
Before Fine-Tuning ⌚ 3.91s | 🧠 9602 MB

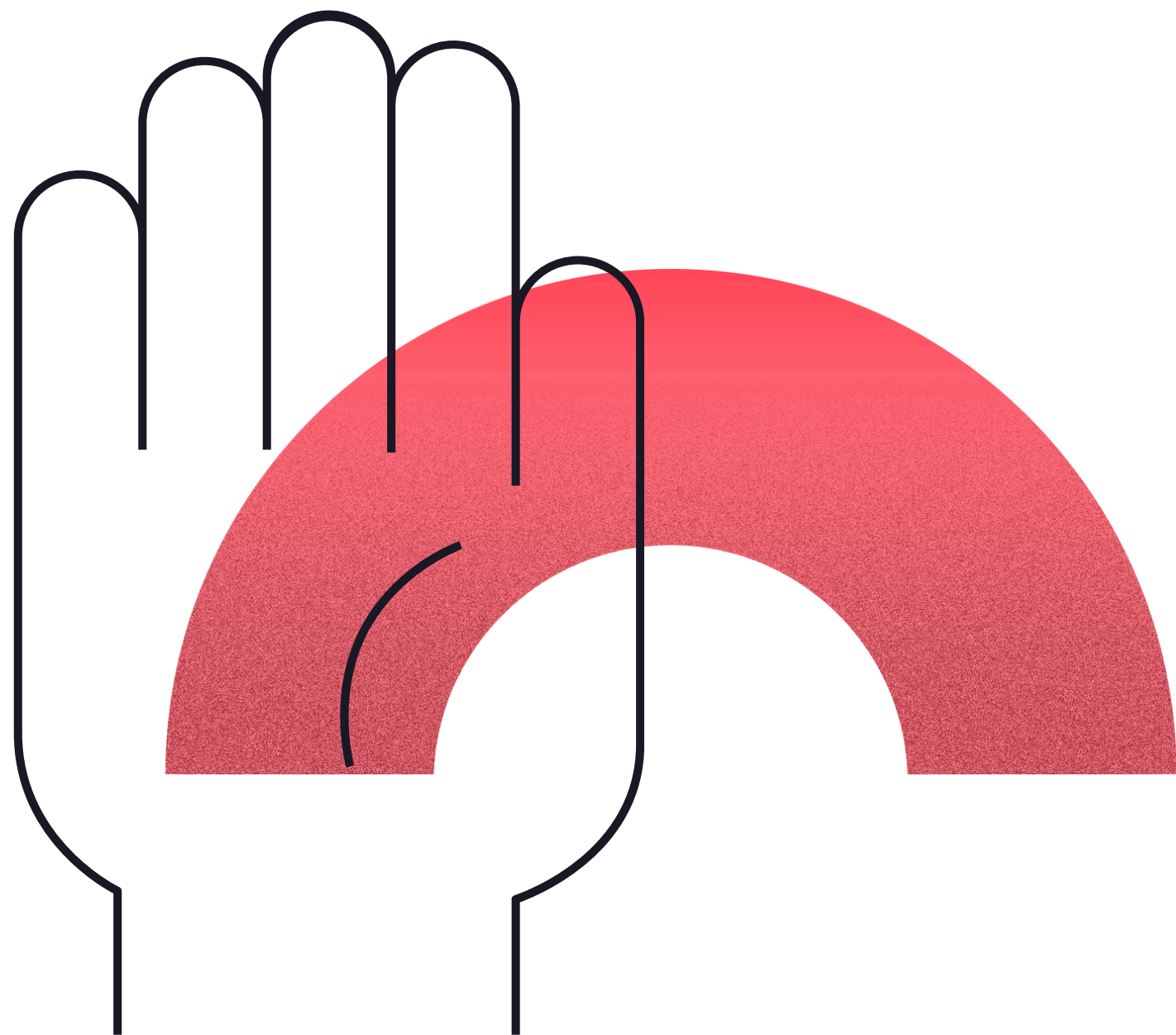


After Fine-Tuning ⌚ 3.87s | 🧠 9602 MB

Evaluation

Metric	Before Fine-Tuning	After Fine-Tuning	Change	Interpretation
CLIP Score	23.55	24.763	1.213	Improved text-image alignment
LPIPS (vs. Reference)	0.781	0.709	-0.072	Improved style similarity to Naruto
FID Score	-	412.527	-	Higher than ideal, suggesting room for improvement
Inception Score	nan	nan	-	Calculation failed, potentially as two models generated are vary





**Thank
you**